

## 1. Product Description

The Allure™ EC-Smart-Vue series sensors are designed to interface with any ECB or ECL-Series controller to provide precision local temperature sensing, information display of system status, and a variety of control functions that can be accessed by room occupants.

Through its user-friendly interface, occupants can view and adjust environmental settings to their liking, for example, view the space temperature, adjust the setpoint, and apply occupancy overrides with the Allure EC-Smart-Vue room temperature sensor. In addition to this, the Allure EC-Smart-Vue-H sensor can measure the room's humidity level, the Allure EC-Smart-Vue-M sensor has a motion activated occupancy sensor, the Allure EC-Smart-Vue-C sensor has a CO<sub>2</sub> sensor to measure air-quality, the Allure EC-Smart-Vue-CM sensor has both an occupancy and CO<sub>2</sub> sensor, the Allure EC-Smart-Vue-CH sensor has both a humidity and CO<sub>2</sub> sensor, the Allure EC-Smart-Vue-HM sensor has a humidity sensor with a built-in motion detector, and the Allure EC-Smart-Vue-CHM sensor has a humidity and CO<sub>2</sub> sensor with a built-in motion detector.

With the EC-gfxProgram programming tool, you can create your own tailor-made display control features that make full use of all Allure EC-Smart-Vue's capabilities. For example, you can program the display to give users feedback on their setpoint selection with the ECO-Vue™ icon that shows more leaves for a setting that not only cares for the environment, but one that also reduces operating costs. A five-character alphanumeric display is available for showing messages.

A password protected technician mode allows an installer to perform commissioning and troubleshooting. When connected with the ECB-VAV or ECL-VAV series, commissioning can start immediately after installation, as the Allure EC-Smart-Vue sensor can be used as a hand-held tool to select the appropriate controller application for the type of HVAC equipment to be controlled, to perform air balancing of the system without requiring an onsite controls engineer, and to troubleshoot the system.

Mounting hardware with a separate sub-base is provided with the device for installation on dry wall or on an electrical junction box. Electrical connection is made through standard Category 5e (Cat 5e) structured cabling.

This cable also brings power and BAS Local Area Network (LAN) signals to the sensor. The LAN is made available at the externally-accessible communications jack, when enabled. By connecting a BACnet MS/TP portable router or a LONWORKS® Network Interface, it is possible to access the LAN for installation and maintenance purposes.

This document describes the hardware installation procedures for the following device models: Allure EC-Smart-Vue, EC-Smart-Vue-H, EC-Smart-Vue-C, EC-Smart-Vue-CH, EC-Smart-Vue-M, EC-Smart-Vue-CM, EC-Smart-Vue-HM, and EC-Smart-Vue-CHM.



Unless otherwise indicated, the term Allure EC-Smart-Vue is used in this document to represent all Allure EC-Smart-Vue models.



Figure 1-1: Allure EC-Smart-Vue, Allure EC-Smart-Vue-H/-C/-CH sensors



Figure 1-2: Models equipped with a motion sensing window in the upper left corner: Allure EC-Smart-Vue-M/-CM/-HM/-CHM sensors

## 2. Cleaning

Clean the Allure EC-Smart-Vue sensors by polishing with a soft dry cloth.

## 3. General Installation Requirements

For proper installation and subsequent operation of the Allure EC-Smart-Vue sensors, pay special attention to the following recommendations:

- It is recommended that the sensor(s) be kept at room temperature for at least 24 hours before installation to allow any condensation that may have accumulated due to low temperature during shipping/storage, to evaporate.
- Upon unpacking the product, inspect the contents of the carton for shipping damages. **Do not install damaged sensors.**
- The device is designed to operate under environmental conditions that are specified in its datasheet.

- Ensure proper ventilation of device and avoid areas where corroding, deteriorating or explosive vapors, fumes or gases may be present.
- Do not drop the sensor or subject it to physical shock.
- If the device is used and/or installed in a manner not specified by Distech Controls, the functionality and the protection provided by the device may be impaired.
- To ensure reading accuracy, handle sensor with care and avoid installing on a vibrating surface.



Make connections to the Allure EC-Smart-Vue sensors last. Yanking the cable while it is attached to the sensor can damage the connector.



Any type of modification to any Distech Controls product will void the product's warranty.



Take reasonable precautions to prevent electrostatic discharges to the device when installing, servicing or operating it. Discharge accumulated static electricity by touching one's hand to a securely grounded object before working with the device.

#### 4. Device Dimensions

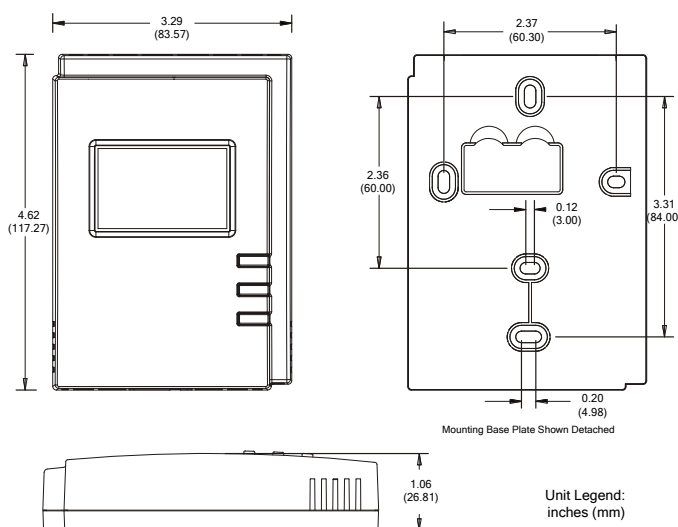


Figure 4-1: Front view, back plate, and side view for models **WITHOUT** motion sensor

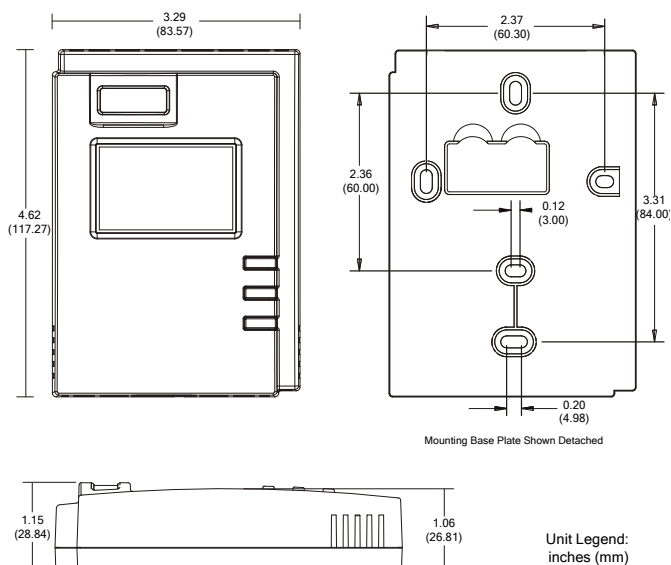


Figure 4-2: Front view, back plate, and side view for models **WITH** motion sensor. Note the motion detection window in the upper left corner of the sensor.

#### 5. General Wiring Recommendations



Turn off power before any kind of servicing.

- All wiring must comply with national and local electrical codes.

#### 6. Mounting Instructions

The Allure EC-Smart-Vue sensors have been specially designed for easy installation. However, certain conditions apply when choosing a suitable location for the device:

- The device should not be installed on an exterior wall.
- The device should not be installed near a heat source.
- The device should not be installed near an air discharge grill.
- The device should not be installed in a place where it can be affected by the sun.
- Install the device in an area that provides proper device ventilation. Nothing must restrain air circulation to the device.
- Models equipped with a motion sensor should be installed in a location and at a height where people will pass in front of it, within its detection zone. See the datasheet for this information.
- Models equipped with a CO<sub>2</sub> sensor must be used in spaces that are periodically unoccupied (e.g. during evening or nighttime hours).



The Allure EC-Smart-Vue sensor has not been designed for outdoor use.

#### Installation Procedure

1. Remove the security screw from the device (Figure 7-1).
2. Open the device by pressing in the two (2) tabs on the bottom of the device with a flat screwdriver and pulling the bottom side of the front plate out.
3. Pull all cables 6" (15 cm) out of the wall, and insert them through the central hole of the back plate.
4. Align the back plate with the wall and mark the location of the mounting holes on the wall. Make sure to orient the proper side of the back plate facing upwards.
5. Remove the back plate and drill holes in the wall if necessary.
6. Install anchors in the wall if necessary.
7. Make sure that the mounting surface is flat and clean.
8. Screw the back plate onto the wall. Do not over tighten.
9. Plug the wire(s) into the connector(s).
10. Gently push excess wiring back into the wall.
11. Set any jumpers. See [Figure 10-1](#).
12. Reattach the front plate and make sure it clips tightly into place. Start by hooking the top in place, and then clip the bottom edge into place.
13. Install security screw.

## 7. Device Components

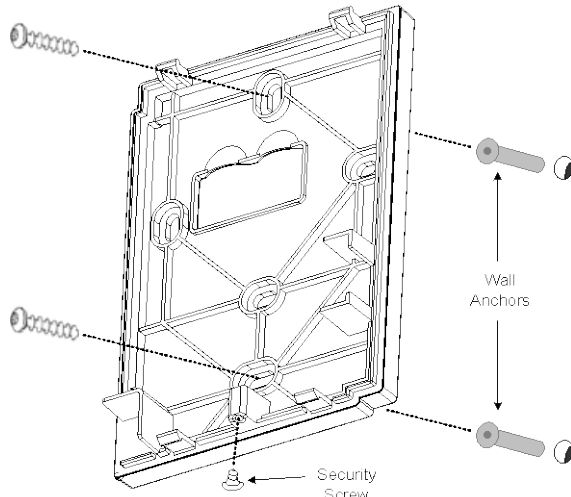


Figure 7-1: Mounting an Allure EC-Smart-Vue

## 8. Supported Quantity

Each controller supports a maximum number of Allure EC-Smart-Vue sensors. The Subnet ID of all Allure EC-Smart-Vue sensors must be set to be within the shown addressing range.

Table 8-1: Number of Allure EC-Smart-Vue sensors supported by controller model

| Series                                                                                     | Maximum Number of Allure EC-Smart-Vue sensors | Permitted Subnet ID Addressing Range |
|--------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------|
| ECB-VAV Series<br>ECL-VAV Series<br>ECB-103<br>ECL-103<br>ECB-203<br>ECL-203               | 4 <sup>1</sup>                                | 1 to 4                               |
| ECB-300<br>ECL-300<br>ECB-400 Series<br>ECL-400 Series<br>ECB-600 Series<br>ECL-600 Series | 12 <sup>1</sup>                               | 1 to 12                              |

<sup>1</sup>A controller can support a maximum of two (2) Allure EC-Smart-Vue models equipped with a CO<sub>2</sub> sensor. The remaining connected Allure EC-Smart-Vue models must be without a CO<sub>2</sub> sensor.

## 9. About the Allure EC-Smart-Vue CO<sub>2</sub> Sensors

The Allure EC-Smart-Vue-CI-CM/CH/CHM are factory calibrated to accurately read CO<sub>2</sub> concentration levels at sea level. When these Allure EC-Smart-Vue sensors are used in locations where the elevation is greater than 500ft (152m) above sea level, you must set the Elevation input of the corresponding ComSensor block in EC-gfxProgram to the current location's elevation to obtain the most accurate readings.

For more information, refer to the Inputs and Outputs section in the EC-gfxProgram User Guide



Under normal conditions, an Allure EC-Smart-Vue sensor with CO<sub>2</sub> sensor will reach its operational accuracy after 25 hours of continuous operation on condition that it was exposed to ambient air reference levels of 400 ppm ±10 ppm CO<sub>2</sub>.

## 10. Connector and Configuration Jumper Location and Identification

Allure EC-Smart-Vues sensors have the following onsite configurable jumpers.

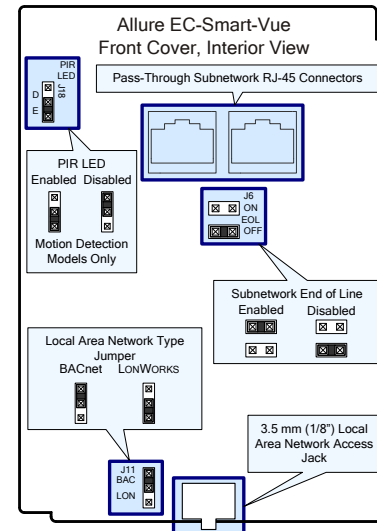
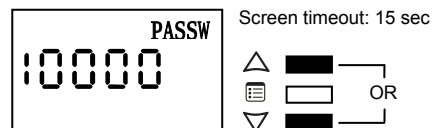


Figure 10-1: Connector and Jumper Locations

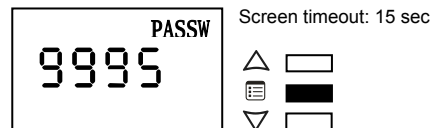
## 11. Setting Up the Allure EC-Smart-Vue with Motion Sensors

The following procedure describes how to commission and test the Allure EC-Smart-Vue sensors equipped with motion sensors as well as adjust the motion sensitivity (Allure EC-Smart-Vue-MI-CM/-HM/-CHM).

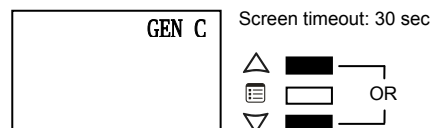
1. Connect an Allure EC-Smart-Vue with motion sensor to the controller with a Cat 5e patch cable.
2. After a 30-second warm-up period, if movement is detected, the motion sensor light will turn on.
3. To test the sensor functionality, aim the sensor in the direction you want to detect motion and walk at a normal pace across the typical detection zone and verify that the sensor light turns on. The light will turn on and after 2 seconds of non-movement, the motion sensor light will turn off.
4. Now to set the motion sensitivity, on the Allure EC-Smart-Vue sensor, press and hold the **Menu** button  for 5 seconds to enter the password menu. 10000 is shown on the display.



5. Press the down button  to set the number to 9995 (this is the default password).

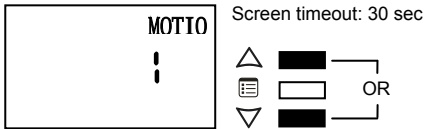


6. Press the **Menu** button  to submit the password. Upon submitting the password, the **GEN CFG** menu appears on the display.



7. To enter the **GEN CFG** submenu, press the down button  once.

8. Press the **Menu** button  several times until **MOTION SENSITIVITY** appears on the display. The current motion sensitivity level is shown.



9. Use the up and down buttons   to set the motion sensitivity level to 0 (low), 1 (medium - typical default setting), or 2 (high). The default and recommended level of sensitivity is medium (1) and should typically work with all installations. This setting should not be changed unless for some reason you are experiencing false detections, then a low sensitivity (0) setting may be used, or if working with larger room installations a high sensitivity (2) setting can be used.
10. Press the Menu button  once to apply the value.
11. Press and hold the Menu button  for 5 seconds to exit the configuration menu.

 The motion indicator LED provides a visual confirmation of motion detection. By default, the indicator (PIR LED jumper) is enabled. If you wish to disable it, set jumper J18 to Disabled. See [Figure 10-1](#).

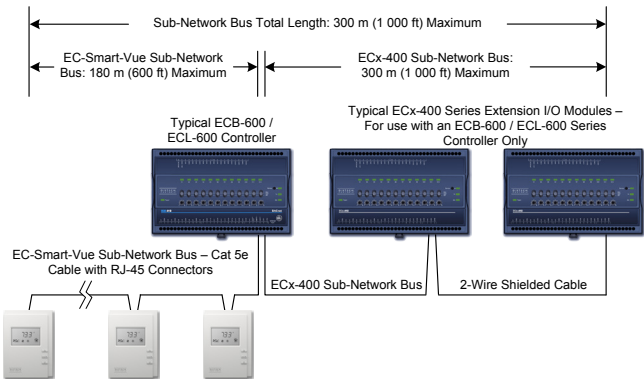
## 12. About the Subnetwork Bus

The ECB-600 and ECL-600 controllers use the subnetwork bus to support the ECx-4XX Series IO Extension Modules through 2-wire shielded cable.

All Distech Controls ECB series and ECL series controllers also use the subnetwork bus to support one or more Allure EC-Smart-Vue sensor(s) using standard structural cabling. See [Table 8-1](#). For the Allure EC-Smart-Vue sensor, the subnetwork bus also extends the LAN to the Allure EC-Smart-Vue room sensor to provide convenient network access for maintenance and troubleshooting purposes.

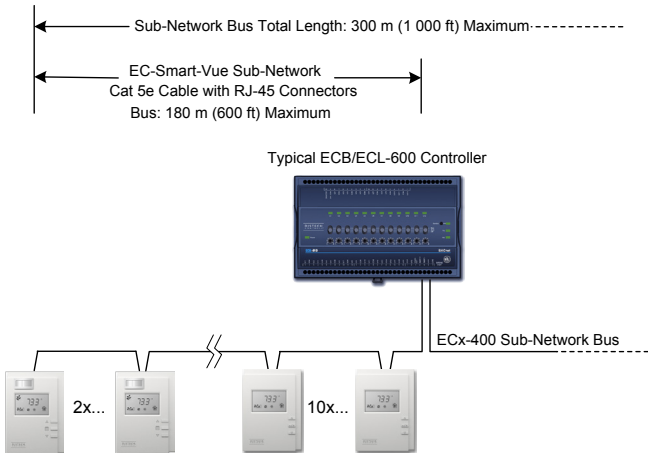
## 13. Subnetwork Bus Total Length

The total maximum length of all subnetwork buses, including both the length of the Allure EC-Smart-Vue subnetwork bus and the ECx-4XX subnetwork bus is 1000 ft (300 m). The maximum length of the Allure EC-Smart-Vue subnetwork bus is 600 ft (180 m). The maximum length of the ECx-4XX subnetwork bus is 1000 ft (300 m).



**Figure 13-1** Subnetwork Bus Overview Showing the Allure EC-Smart-Vue Subnetwork Bus and the ECx-4XX Subnetwork Bus

A controller can support a maximum of two (2) Allure EC-Smart-Vue models equipped with a CO<sub>2</sub> sensor; the remaining connected models must be without a CO<sub>2</sub> sensor. For instance, if the subnetwork can consist of 12 Allure EC-Smart-Vue sensors in all, then 10 sensor models will be without a CO<sub>2</sub> sensor and the other two (2) will be equipped with a CO<sub>2</sub> sensor. To ensure proper functioning, it is recommended to distribute the sensors throughout the subnetwork.



**Figure 13-2** Allure EC-Smart-Vue Sensors Subnetwork Length and Distribution

## 14. Connection Cable

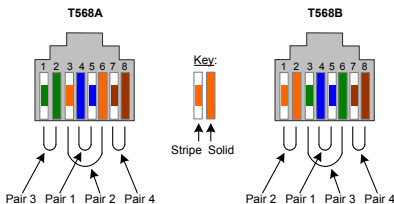
Connect the Allure EC-Smart-Vue sensor to the controller with a standard Cat 5e Ethernet patch cable fitted with RJ-45 connectors.

 If you make your own patch cable, use Cat 5e cable and crimp the RJ-45 connectors at both ends of the cable either as T568A or T568B.

**Table 14-1:** T568A and T568B Terminations for an RJ-45 Connector

| Pin | T568A (at both cable ends) |                     | T568B (at both cable ends) |                     |
|-----|----------------------------|---------------------|----------------------------|---------------------|
|     | Pair                       | Color               | Pair                       | Color               |
| 1   | 3                          | white/green stripe  | 2                          | white/orange stripe |
| 2   | 3                          | green solid         | 2                          | orange solid        |
| 3   | 2                          | white/orange stripe | 3                          | white/green stripe  |
| 4   | 1                          | blue solid          | 1                          | blue solid          |
| 5   | 1                          | white/blue stripe   | 1                          | white/blue stripe   |
| 6   | 2                          | orange solid        | 3                          | green solid         |
| 7   | 4                          | white/brown stripe  | 4                          | white/brown stripe  |
| 8   | 4                          | brown solid         | 4                          | brown solid         |

The final result of a crimped RJ-45 connector is shown graphically below.



**Figure 14-1:** T568A and T568B Crimp Wire Sequence for an RJ-45 Connector

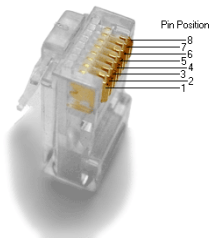


Figure 14-2: Pins on RJ-45 Jack Face

Patch cables fitted with connectors supplied by Distech Controls are wired as T568B.

## 15. Subnetwork Bus Topology and EOL Terminations

Only a daisy-chain topology is acceptable for the EC-Smart-Vue subnetwork Bus. T-connections are not allowed.

When one or more Allure EC-Smart-Vue sensors are installed with an ECB-VAV Series, ECB-103, ECB-203, ECB-300, ECB-400 Series, ECL-VAV Series, ECL-103, ECL-203, ECL-300, or ECL-400 Series Controller, only the EOL terminations of the last Allure EC-Smart-Vue sensor are set to ON. All other Allure EC-Smart-Vue sensors must have their EOL terminations set to OFF. The controller must be the first device on the bus as the EOL termination in these devices are permanently enabled.



A controller can support a maximum of two (2) Allure EC-Smart-Vue models equipped with a CO<sub>2</sub> sensor. The remaining connected Allure EC-Smart-Vue models must be without a CO<sub>2</sub> sensor.

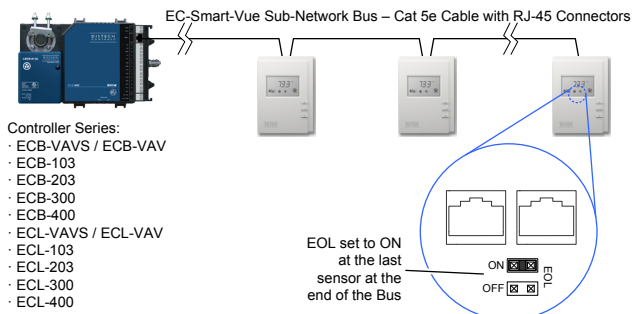


Figure 15-1: Setting the EOL Terminations on the Allure EC-Smart-Vue Subnetwork Bus for non ECB-600 or ECL-600 series controllers

When one or more Allure EC-Smart-Vue sensors are installed with an ECB-600 / ECL-600 (without an ECx-4XX IO Extension Module), only the EOL terminations on the ECB-600 / ECL-600 and the last Allure EC-Smart-Vue sensor are set to ON. All other Allure EC-Smart-Vue sensors must have their EOL terminations set to OFF.

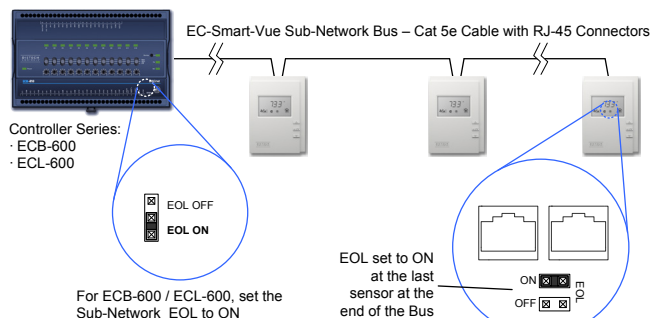


Figure 15-2: Setting the EOL Terminations on the Allure EC-Smart-Vue Subnetwork Bus for ECB-600 or ECL-600 Series controllers

When ECx-400 Series IO Extension Modules are installed with an ECB-600 / ECL-600 Series controller and with Allure EC-Smart-Vue sensor(s), only the EOL terminations on the last ECx-400 Series IO Extension Module and the last EC Smart Vue are set to ON. All other ECx-400 Series IO Extension Modules and Allure EC-Smart-Vue models must have their EOL terminations set to OFF.

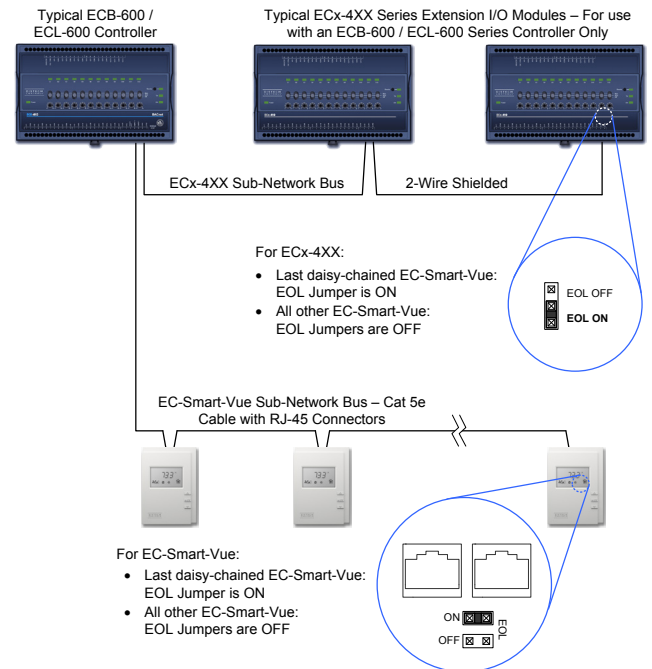


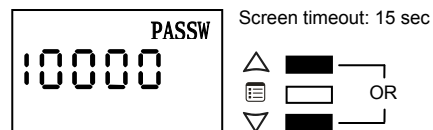
Figure 15-3: Setting the EOL Terminations when Allure EC-Smart-Vue Sensors are used with ECx-400 Series IO Extension Modules

ECx-400 Series devices and Allure EC-Smart-Vue sensors are factory-set with the EOL set to OFF by default.

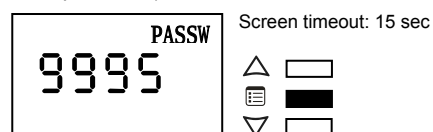
## 16. Setting the Allure EC-Smart-Vue Subnet ID

Controllers can be commissioned with an Allure EC-Smart-Vue sensor. The default Subnet ID for an Allure EC-Smart-Vue is 1. To commission a controller, the Allure EC-Smart-Vue's Subnet ID must be set to 1. If the Allure EC-Smart-Vue's Subnet ID has been set to another value (for example, the display flashes error code 1 with the Bell icon when the Allure EC-Smart-Vue sensor is connected to a controller for commissioning), change the Subnet ID to 1 as follows:

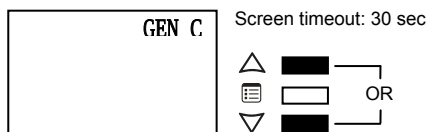
1. Connect an Allure EC-Smart-Vue sensor to the controller with a Cat 5e patch cable. Wait for the Bell icon and the number 1 to flash on the display.
2. Press and hold the Menu button for 5 seconds to enter the password menu. 10000 is shown on the display.



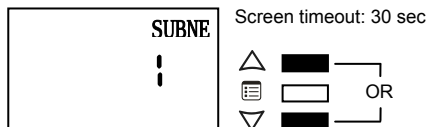
3. Press the down button to set the number to 9995 (this is the default password).



4. Press the Menu button to submit the password. Upon submitting the password, the GEN CFG menu appears on the display.



- Press the down button once to enter the **GEN CFG** submenu.
- Press the Menu button several times until SUBNET ID appears on the display. The current controller's Subnet ID is shown.



- Use the up and down buttons to set the controller's Subnet ID to 1. *Tip:* Hold down either the up or down button to fast-advance the display value.
- Press the Menu button once.
- Press and hold the Menu button for 5 seconds to exit the configuration menu.

The Allure EC-Smart-Vue sensor can now be used to go from one controller to the next for commissioning purposes.

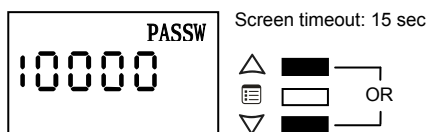
## 17. Commissioning Controllers

To commission a controller with a MAC Address DIP switch located on the faceplate or under the cover in the case of an ECB-VAV Series controller, first set the DIP switch to 0 (all off). When using an Allure EC-Smart-Vue sensor for commissioning a controller (before code is downloaded to the controller from EC-gfxProgram), connect an Allure EC-Smart-Vue sensor to the controller with its Subnet ID set to 1. (see [Section 16](#)).

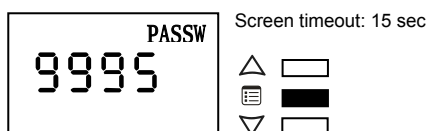
For ECB-VAV and ECL-VAV Series controllers, commissioning can be used to perform application selection if needed. Applications are pre-loaded programs that enable the ECB-VAV and ECL-VAV to control a typical VAV box. See the [Pre-Loaded Application User Guide](#) for more information.

For ECB Series controllers only, during commissioning, the Allure EC-Smart-Vue sensor is used to set the controller's BACnet® MAC Address. Set the connected ECB Series controller's MAC Address as follows:

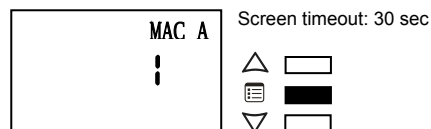
- Connect an Allure EC-Smart-Vue sensor to the controller with a Cat 5e patch cable. Wait for the display to show the room temperature.
- Press and hold the Menu button for 5 seconds to enter the password menu. 10000 is shown on the display.



- Use the down button to set the number to 9995 (this is the default password).



- Press the **Menu** button to submit the password. Upon submitting the password, the **GEN CFG** menu appears on the display.
- Press the down button once to enter the **GEN CFG** submenu. The **MAC ADDRESS** menu is shown with the current controller's BACnet MAC Address.



- Use the up and down buttons to set the controller's MAC Address. Only addresses from 1 to 127 are recommended to be used.
- Press the Menu button once to apply the value.
- Press and hold the Menu button for 5 seconds to exit the configuration menu.

Once the controller's communications network is operational, the controller can be programmed with EC-gfxProgram. For each Allure EC-Smart-Vue sensor, set its Subnet ID number to the block number of its associated ComSensor block in EC-gfxProgram. This is done in the Allure EC-Smart-Vue's **GEN CFG** menu under **SUBNET ID**. The range of valid Allure EC-Smart-Vue Subnet IDs is shown in Table 8-1.

## 18. Setting the BAUD Rate (optional – BACnet controllers only)

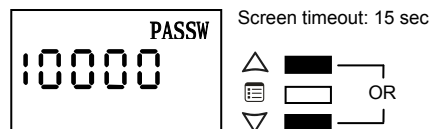
By default, the BAUD rate for the controller is set to automatically detect the current communication BAUD rate of the connected BACnet MS/TP network (AUTO). This is the preferred setting for a controller. However, at least one controller on the BACnet MS/TP network data bus must have its BAUD rate set. The preference is to set the building controller's BAUD rate (if present). Otherwise, set the BAUD rate on one controller that will set the BAUD rate for all other controllers (to act as the master for setting the BAUD rate).



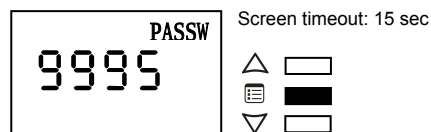
When the Baud rate is set to AUTO, the controller cannot initiate any communication until it has detected the baud rate of the BACnet MS/TP network. If all controllers on the BACnet MS/TP network are set to AUTO, then all controllers will not communicate.

Set the connected controller's BAUD rate as follows:

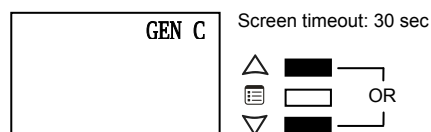
- Connect an Allure EC-Smart-Vue sensor to the controller with a Cat 5e patch cable. Wait for the display to show the room temperature.
- Press and hold the Menu button for 5 seconds to enter the password menu. 10000 is shown on the display.



- Use the down button to set the number to 9995 (this is the default password).

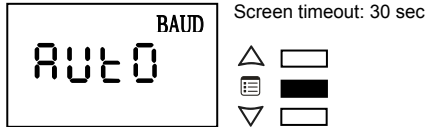


- Press the **Menu** button to submit the password. Upon submitting the password, the **GEN CFG** menu appears on the display.



- Press the down button once to enter the **GEN CFG** submenu.

- Use the Menu button  several times until **BAUD RATE** appears on the display. The current controller's BAUD rate is shown.



- Use the up and down buttons   to set the controller's Baud rate. The **AUTO** setting detects and uses the current baud rate being used by the BACnet MS/TP network.
- Press the Menu button  once to apply the value.
- Press and hold the Menu button  for 5 seconds to exit the configuration menu.

## 19. Set the LAN Type

Set the BAC/LON jumper for the type LAN in use: BAC for a BACnet network, LON for a LONWORKS network.

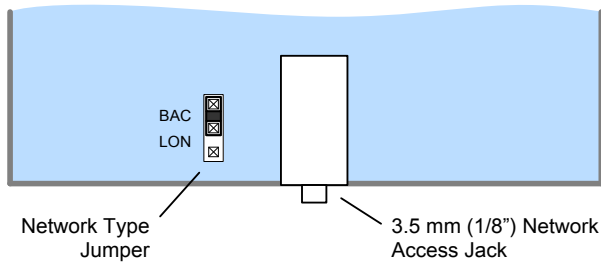


Figure 19-1: BAC/LON Jumper Set to the BAC (BACnet) Position

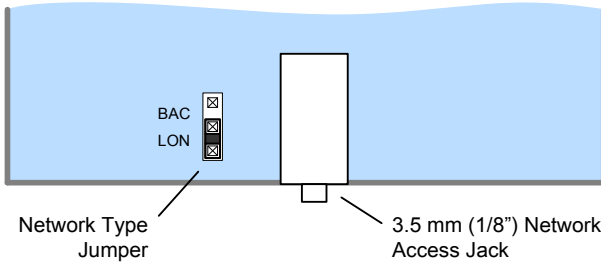


Figure 19-2: BAC/LON Jumper: Set to the LON (LONWORKS) Position

## 20. BACnet Network Access from the Sensor

To temporarily access the BACnet MS/TP LAN for commissioning and maintenance purposes, connect a BACnet MS/TP Adaptor to the audio plug port located on the lower edge of the Allure EC-Smart-Vue sensor. Wire a standard  $\frac{1}{8}$ " (3.5 mm) three-conductor stereo jack as shown below.



The BACnet MS/TP Adaptor must have an electrically-isolated RS-485 port. Otherwise a ground path from the BACnet network will be made through the computer that will disrupt BACnet network communications.

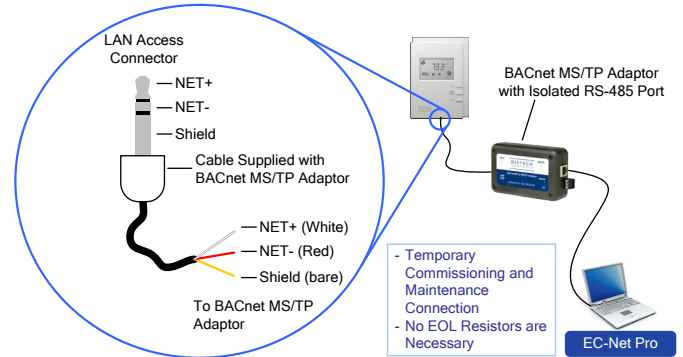


Figure 20-1:  $\frac{1}{8}$ " (3.5 mm) Stereo Jack Connection for a Portable Router

## 21. LONWORKS Network Access from the Sensor

For commissioning and maintenance purposes, the LONWORKS network is optionally available from the Allure EC-Smart-Vue audio plug port. By setting the two (2) **Net to Subnet Port Settings** jumpers inside the ECL series controller to **Enable** will connect the main LONWORKS network to the Allure EC-Smart-Vue subnetwork Cat 5e cable.



**Recommendation:** Only a limited number of controllers on a LONWORKS network segment should have their **Net to Subnet Port Settings** jumpers enabled. Enabling too many Allure EC-Smart-Vue sensors with network access may cause network communication issues. If there are any network communication problems, refer to Section 22.

The Cat 5e cable length is restricted by the maximum allowable subnetwork bus length (see [Subnetwork Bus Total Length](#)). The standard **Net to Subnet Port Settings** jumper setting is **Disable**.

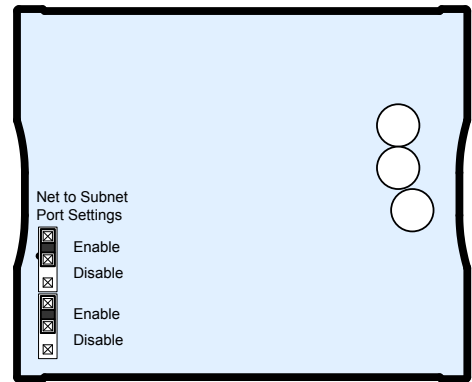


Figure 21-1: ECL series controller: Net to Subnet Port Settings Jumpers

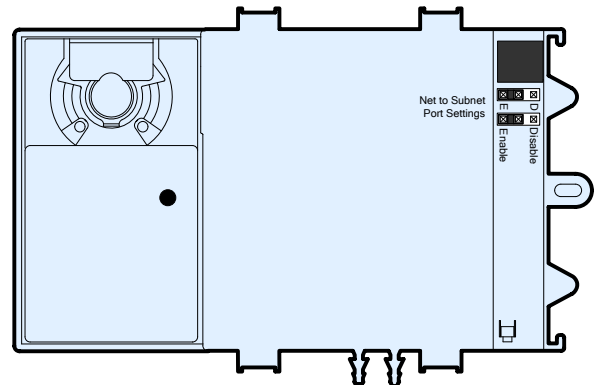


Figure 21-2: ECL-VAV series controller: Net to Subnet Port Settings Jumpers

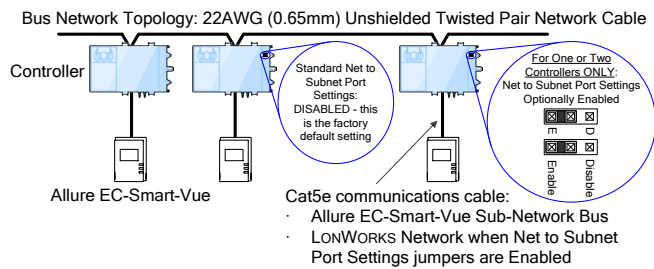


Figure 21-3: LONWORKS Network: Bus Topology

To temporarily access the LONWORKS LAN for commissioning and maintenance purposes, connect a LONWORKS network interface to the audio plug port located on the lower edge of the Allure EC-Smart-Vue sensor. Wire a standard  $\frac{1}{8}$ " (3.5 mm) three-conductor (stereo jack) or two-conductor (mono jack) as shown below.

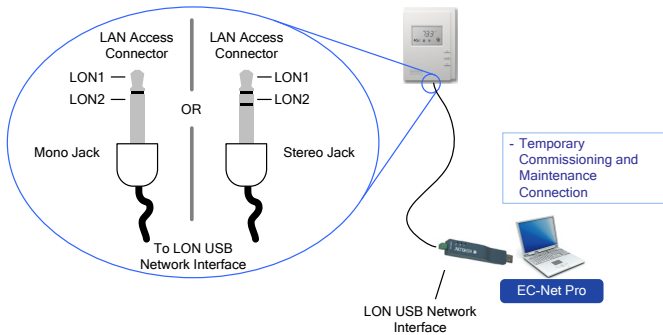


Figure 21-4:  $\frac{1}{8}$ " (3.5 mm) Stereo or Mono Jack Connection for a LONWORKS Network Interface

## 22. Troubleshooting Guide

### Allure EC-Smart-Vue screen is blank & back light is off

|                                                                          |                                                                                                                                                                                                            |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is the Allure EC-Smart-Vue sensor connected to the controller?           | Verify that the Allure EC-Smart-Vue sensor is connected to the controller and that the patch cables are plugged into the connectors. Refer to <a href="#">Installation Procedure</a> for more information. |
| Is power being supplied to the controller?                               | There may be no power being supplied from the controller. Check if the controller has power or if the controller's internal fuses have blown or tripped.                                                   |
| Is the cable connected to the controller and Allure EC-Smart-Vue sensor? | Verify wiring.                                                                                                                                                                                             |
| Was the patch cable made onsite?                                         | Verify that the RJ-45 crimp connectors were installed on the cable correctly. See <a href="#">Connection Cable</a> .                                                                                       |

### Allure EC-Smart-Vue screen is blank & back light is on for about 30 to 45 seconds – Normal Operation

|                              |                                                                                                                                        |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Firmware upgrade in progress | Wait for the upgrade to complete. Do not disconnect the sensor from the controller as the upgrade will restart once it is reconnected. |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|

### Device not communicating with controller

|                                                      |                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is the address correctly set to a unique address?    | Each Allure EC-Smart-Vue sensor must be set to a unique address for each controller. See <a href="#">Setting the Allure EC-Smart-Vue Subnet ID</a> .                                                                                                                                                                                       |
| Is the device too far from controller?               | Verify the distance between the device and the controller. See <a href="#">Subnetwork Bus Total Length</a> .                                                                                                                                                                                                                               |
| Is there a configuration problem?                    | With EC-gfxProgram, check the configuration of the sensor; for example, is it enabled? Refer to the <a href="#">EC-gfxProgram User Guide</a> for more information.                                                                                                                                                                         |
| Have the subnetwork EOL settings been correctly set? | Only the last Allure EC-Smart-Vue sensor must have its EOL termination set to ON. See <a href="#">Figure 15-1</a> and <a href="#">Figure 15-2</a> . When one or more ECx-400 Series IO Extension modules are connected to the controller, only the last ECx-400 must have its EOL termination set to ON. See <a href="#">Figure 15-3</a> . |

### Controller cannot communicate on the LONWORKS network

|                                                                   |                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Too many Allure EC-Smart-Vue sensors are providing network access | Disable the <b>Net to Subnet Port Settings</b> jumpers on all controllers (for jumper location, see <a href="#">Section 21</a> ). If communications are re-established, re-enable only a few Allure EC-Smart-Vue sensors to have network access. |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Allure EC-Smart-Vue motion detector window indicator is always ON and the Motion output of the associated ComSensor block always reads NULL in EC-gfxProgram

|                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does the connected controller have Allure EC-Smart-Vue firmware that supports the motion sensor? | When the Allure EC-Smart-Vue sensor is connected to a controller, its firmware is loaded from the controller. In this case, the controller has an earlier version of Allure EC-Smart-Vue firmware that does not support the motion sensor. To upgrade to the latest Allure EC-Smart-Vue firmware, download the firmware from SmartInstaller and refer to the firmware upgrade procedure in the <a href="#">EC-gfxProgram User Guide</a> . |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### The CO<sub>2</sub> output of the associated ComSensor block always reads NULL in EC-gfxProgram

|                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does the connected controller have Allure EC-Smart-Vue firmware that supports the CO <sub>2</sub> sensor? | When the Allure EC-Smart-Vue sensor is connected to a controller, its firmware is loaded from the controller. In this case, the controller has an earlier version of Allure EC-Smart-Vue firmware that does not support the CO <sub>2</sub> sensor. To upgrade to the latest Allure EC-Smart-Vue firmware, download the firmware from SmartInstaller and refer to the firmware upgrade procedure in the <a href="#">EC-gfxProgram User Guide</a> . |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### The CO<sub>2</sub> sensor readings are too high, too low, or inconsistent between sensors

|                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Immediately after installing the Allure EC-Smart-Vue with CO <sub>2</sub> sensors, are the CO <sub>2</sub> sensor readings incoherent? | <p>If the CO<sub>2</sub> sensor readings seem unusual or show inconsistencies between sensors in the same building right after installation, the following reasons should be taken into consideration:</p> <ul style="list-style-type: none"> <li>• Concentration levels in each space may be different</li> <li>• The installer may have unintentionally blown into the sensor while installing it.</li> <li>• The sensor may have been dropped or mishandled during shipment causing a minor shift in the original factory calibration.</li> </ul> <p>The recommendation is to allow up to 14 days of operation (without power interruptions) for the sensor to calibrate itself according to its new environment.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Error Code Interpretation

|                                                        |                                                                         |                                                                                                                                                                    |
|--------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clock icon flashing for 15 seconds                     | Cannot communicate with controller.                                     | Wait for the communication link to the controller to be established.                                                                                               |
| After 15 seconds: Flashing error code 1 with Bell icon |                                                                         | Verify wiring                                                                                                                                                      |
| Flashing error code 2 with Bell icon                   | Invalid configuration.                                                  | Verify that all Allure EC-Smart-Vue's Subnet IDs are unique for this controller. See <a href="#">Setting the Allure EC-Smart-Vue Subnet ID</a> .                   |
| Flashing error code 3 with Bell icon                   | Allure EC-Smart-Vue sensor is not properly configured in the controller | In EC-gfxProgram, resynchronize the code with the controller. Contact Distech Controls Customer Support.                                                           |
|                                                        |                                                                         | With EC-gfxProgram, check the configuration of the sensor, for example, is it enabled? Refer to the <a href="#">EC-gfxProgram User Guide</a> for more information. |

## Total Quality Commitment

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